

Q6: Declare type: Char
 Declare currentcharge, requiredcharge, duration, discount: real
 Declare currenttime: time
 Declare membership, disable, available, parking: Boolean
 Declare priority, status: String, cprice, pprice, totalprice: INT
 discount, pprice, cprice, totalprice $\leftarrow$ 0, status $\leftarrow$ " ", parking, charge $\leftarrow$ FALSE
 Output "Enter current charge, requiredcharge, duration, currenttime"
 (in 24 hr clock), membership (Yes or No), Disable (Yes or No),
 available (Yes or No), type (E for EV, H for hybrid): "
 Input currentcharge, requiredcharge, duration, currenttime,
 membership, disable, available, type
 If available = False then
 If type = 'E' then
 Output "No charging Slot available" ~~break break~~
 Else if type = 'H' then
 Output "Charging unavailable - parking only"
 Endif ~~input parking~~ input parking
 Endif ~~break~~ AND(available=TRUE)
 If ((type='E') OR (type='H' AND currentcharge < 40)) then
 requiredcharge $\leftarrow$ requiredcharge - currentcharge
 If requiredcharge <= currentcharge then
 Output "no charging required"
 Else if (currentcharge <= 15) AND (required charge >= 80) then
 priority $\leftarrow$ "Emergency Charging Priority"
 Else if (membership = TRUE OR disable = TRUE) AND
 currentcharge <= 30 then
 priority $\leftarrow$ "Priority charging"
 Else

priority $\leftarrow$ "Normal charging"

EndIf

EndIf

EndIf

If currenttime $<$ "17:00:00" OR currenttime $>$ "22:00:00" then
 status $\leftarrow$ "off" $\rightarrow$ If member = ~~TRUE~~ AND priority $\neq$ "Emergency charging Priority" then
 $cprice \leftarrow 35 \times 0.80$ discount $\leftarrow cprice \times 0.20$

Else

$cprice \leftarrow 35$

discount $\leftarrow 0$

EndIf

Else If currenttime $\geq$ "17:00:00" AND currenttime $\leq$ "22:00:00"

then

status $\leftarrow$ "peak" $\rightarrow$ If member = ~~TRUE~~ AND priority $\neq$ "Emergency charging Priority" then
 $cprice \leftarrow 50 \times 0.90$ discount $\leftarrow cprice \times 0.10$

Else

$cprice \leftarrow 50$, discount $\leftarrow 0$

EndIf

EndIf

EndIf

~~charge~~ $\leftarrow$ TRUE charge $\leftarrow$ TRUE

$cprice = \text{charging required} \times cprice$, discount $\leftarrow$ discount $\times$ charging required

If parking = TRUE OR charge = TRUE

If duration ≤ 2 then

$pprice \leftarrow 200$

Else if duration > 2 AND duration ≤ 5 then

$pprice \leftarrow 400$

Else if duration > 5 then

$pprice \leftarrow 700$

EndIf

EndIf

Else
 Output "Vehicle does not
 qualify for EV
 charging"
 EndIf

EndIf

If membership = TRUE then

pprice $\leftarrow$ pprice * 0.80

discount $\leftarrow$ discount + (pprice * 0.20)

Else if disable = TRUE then

pprice $\leftarrow$ 0

discount $\leftarrow$ discount + pprice

EndIf

EndIf

EndIf ~~EndIf~~

totalprice $\leftarrow$ cprice + pprice

If duration > 8 then

Output "long-stay coming; please relocate your vehicle after charging"

Else

Output "Standard parking duration"

EndIf

If parking = TRUE then

Output ("Parking price: ", totalprice)

Else

Output type, currentcharge, requiredcharge, priority, status, cprice, pprice, discount, totalprice.

Q6: Flowchart:





